

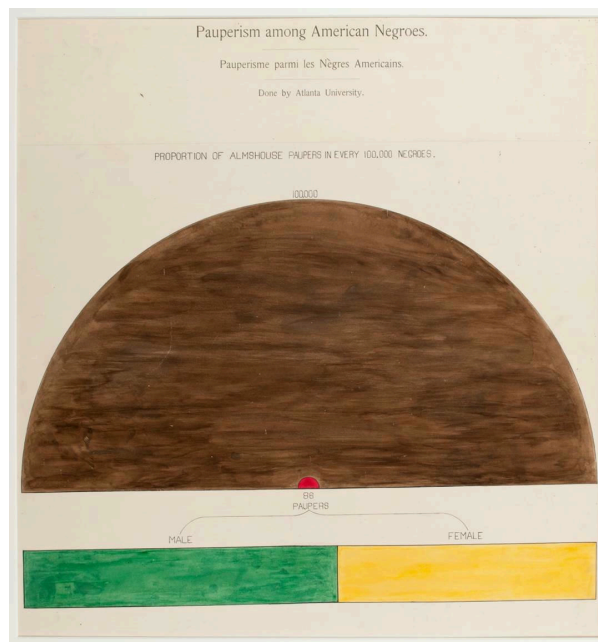
ECON345 Homework 8: Advanced Plots and Regression I

Due: October 29th, 11:59 PM

Develop your code in an R script first if that helps you think, then answer the questions (including your code) in an R Markdown file. Comment your code to explain what you're doing – partial credit rewards a reasonable, well-documented attempt, not just a correct final answer. For any part you can't produce output for, briefly explain what you tried and what went wrong.

Part 1: Visualizing to Communicate (10 points)

Read [“W. E. B. Du Bois’ staggering Data Visualizations are as powerful today as they were in 1900”](#) and [“Part 4: Style and Rich Detail: On Viewing an Original W.E.B. Du Bois Data Visualization”](#), then consider the following chart on “Pauperism Among American Negroes.”



Question 1.1 (5 points)

According to the reading, who was the audience for the exhibit, and what was the purpose of this chart?

Question 1.2 (5 points)

Which items on the data visualization checklist (distributed with Lecture 7) does this chart follow? What is its main message, and how effectively do you think it conveys that message?

Part 2: The Grammar of Graphics, Beautified (30 points, +10 bonus)

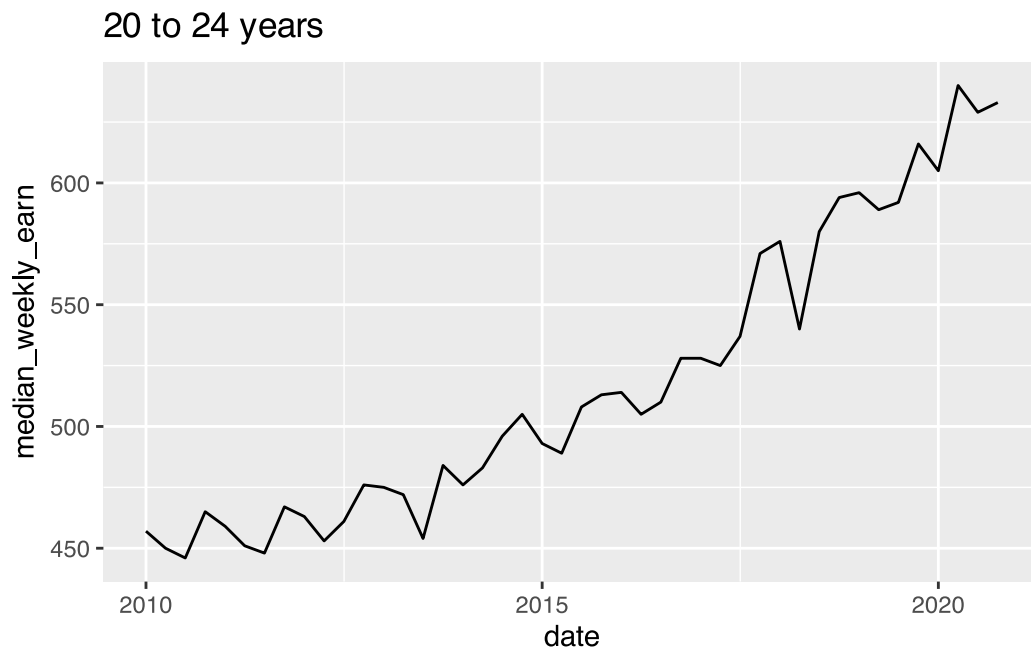
This part uses `median_earnings.csv`, from the Bureau of Labor Statistics, which has median weekly earnings by age group over time.

```
glimpse(median_earnings)
```

```
Rows: 352
Columns: 3
$ date          <date> 2010-01-01, 2010-04-01, 2010-07-01, 2010-10-01,
20...
$ age           <chr> "16 to 19 years", "16 to 19 years", "16 to 19
years...
$ median_weekly_earn <dbl> 356, 346, 335, 360, 360, 349, 356, 346, 362, 357,
3...
```

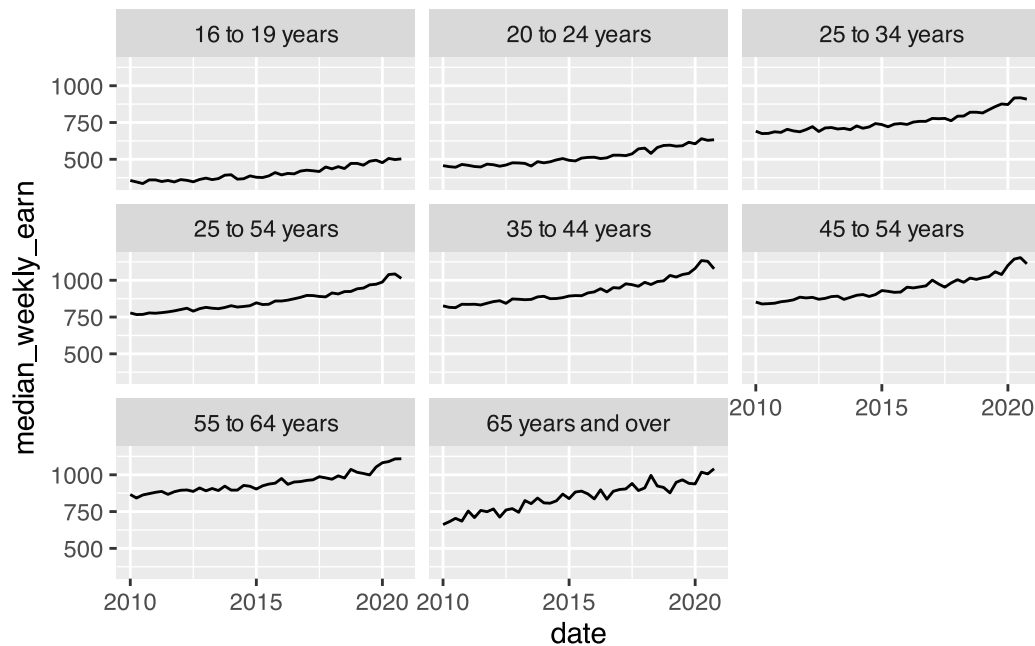
Question 2.1 (6 points)

Create a separate line plot for each of the 8 age groups, showing how median earnings evolved over time. Copying and pasting the same plotting code 8 times is exactly the repetition a loop should replace — use a `for` loop over the distinct age groups, filtering to each one in turn. (Reminder: inside a loop, you must explicitly `print()` the plot — just calling the plot object does nothing.) One plot should look like:



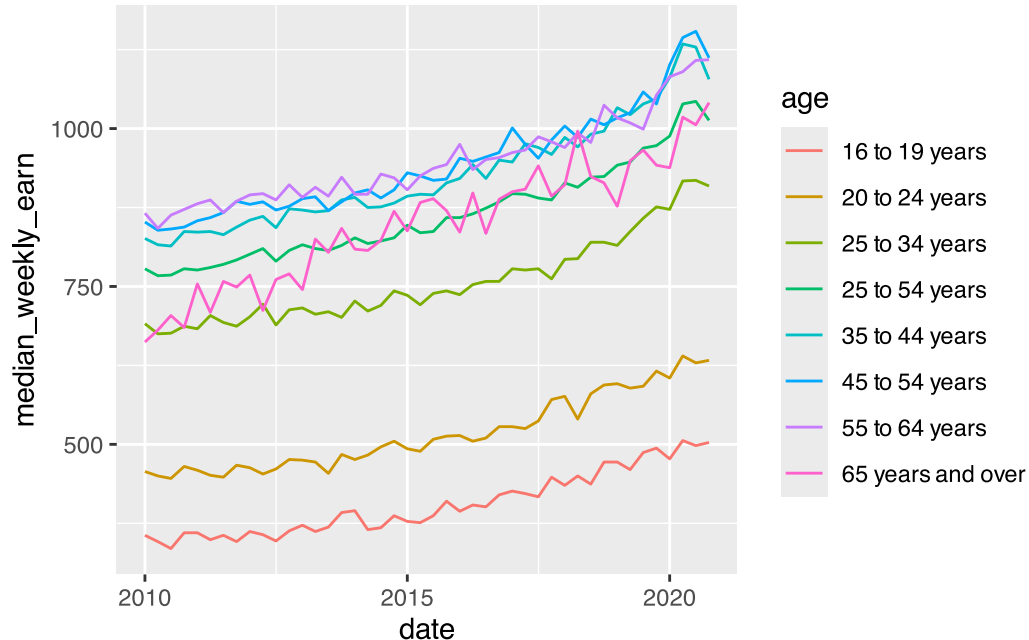
Question 2.2 (6 points)

Eight separate plots makes comparison difficult. Using `facet_wrap()`, put every age group on one figure, each in its own panel:



Question 2.3 (5 points)

Faceting is still hard to compare at a glance. Using the Grammar of Graphics, put every age group's **line** on the same **plot** instead:



Question 2.4 (7 points)

Beautify the plot from 2.3 with `labs()`:

- A title and subtitle that support the plot's main message, and a caption listing the data source
- Titles for the x and y axes
- No title on the color legend, since your title/subtitle already makes clear it's age groups

Question 2.5 (3 points)

Format the y-axis tick labels as dollar amounts with a `scale_*` function.

Question 2.6 (3 points)

Add a pre-made theme of your choice, then use `theme()` to left-align the title and italicize the subtitle.

Bonus (10 points)

`earn_by_race_sex.csv` tracks median weekly earnings over time by race and sex instead of age group:

```
glimpse(earn_by_race_sex)
```

```

Rows: 66
Columns: 5
$ year          <dbl> 2010, 2010, 2010, 2010, 2010, 2010, 2011, 2011,
201...
$ sex           <chr> "Men", "Men", "Men", "Women", "Women", "Women",
"Me...
$ race          <chr> "Asian", "Black or African American", "White",
"Asi...
$ median_weekly_earn <dbl> 943.5, 655.5, 875.0, 767.0, 604.5, 711.5, 976.0,
67...
$ date          <date> 2010-01-01, 2010-01-01, 2010-01-01, 2010-01-01,
20...

```

Make any plot you like that visualizes this data, following the Grammar of Graphics, using **more than just the x and y aesthetics** — color, linetype, fill, alpha, or anything else. Beautifying it further (title, subtitle, theme) is encouraged but not required. You'll get full credit for any choice of plot as long as you describe, in words, which variable is mapped to which aesthetic and which geometry (or geometries) you used.

Part 3: A First Regression (30 points)

This part returns to the Redfin home sales data from lecture (`redfin_clean.csv`).

Question 3.1 (5 points)

Using `vtable::st()` (or `kable()`), display summary statistics for price, baths, and square_feet, with readable variable names. (See [summary_tables_with_vtable_package.pdf](#), distributed with Lecture 4, for an example.)

```

df <- redfin %>%
  select(`Price (USD)` = price, `Bathrooms` = baths, `Square Feet` =
square_feet)

st(df, group = NA, out = "return", add.median = TRUE, digits = 1) %>%
  knitr::kable(align = 'l', caption = "Summary of Redfin Variables")

```

Variable	N	Mean	Std. Dev.	Min	Pctl. 25	Pctl. 50	Pctl. 75	Max
Price (USD)	1173	620407	462286	1490	330000	490000	784000	4550000
Bathrooms	1146	2	1	1	1	2	3	8
Square Feet	1161	1607	1100	252	899	1296	1922	8364

Question 3.2 (5 points)

Regress price (dependent variable) on baths (explanatory variable) with `lm()`, and display the results with `broom::tidy()` — **not** `summary()` or `coef()`. Verify your output matches:

```
price_baths_mod <- lm(price ~ baths, data = redfin)

tidy(price_baths_mod) %>%
  mutate(p.value = scales::pvalue(p.value)) %>%
  knitr::kable(digits = 2)
```

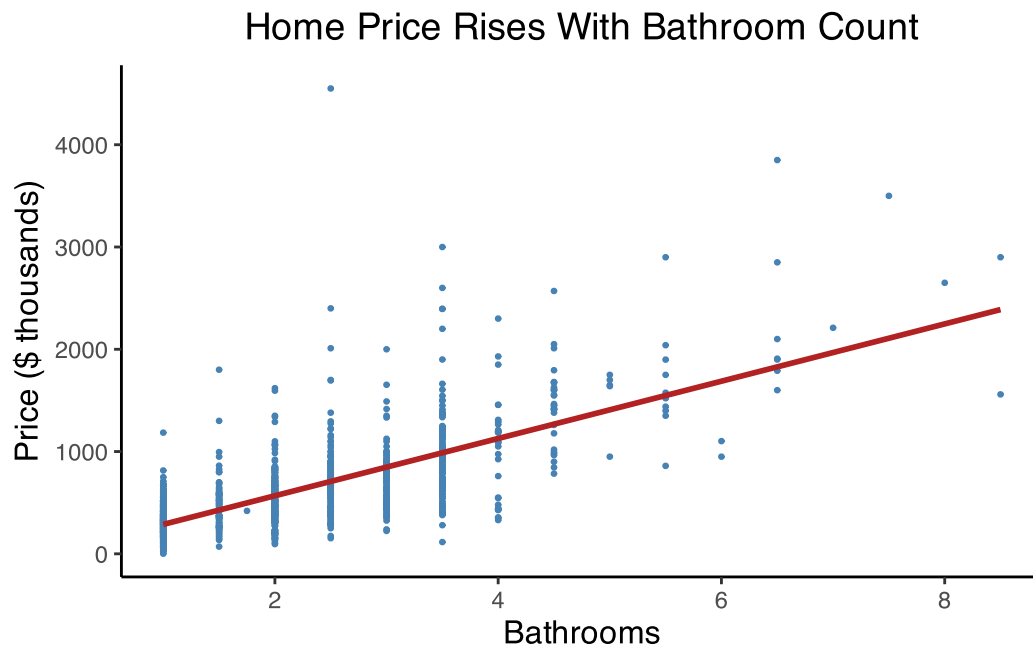
term	estimate	std.error	statistic	p.value
(Intercept)	7551.21	20391.94	0.37	0.711
baths	280058.28	8192.13	34.19	<0.001

Question 3.3 (10 points)

Visualize the regression by (1) making a scatterplot of price against baths and (2) adding the fitted line from your regression using `broom::augment()` — not `coef()` and `geom_abline()`. Your plot should also:

- Have a descriptive title
- Have clear, informative axis labels, in a larger font than the default
- Drop the grid lines and gray background, while keeping the x- and y-axis lines (hint: `theme()` elements — see the lecture slides)

Your output should look like:



Question 3.4 (4 points)

In words, what does the coefficient on baths mean? Be specific about units.

Question 3.5 (3 points)

Interpret your results in terms of both statistical and economic significance. (Check the summary statistics from 3.1 before judging magnitude.)

Question 3.6 (3 points)

Look again at your scatterplot. Does a straight line seem like an appropriate summary of this relationship across the whole range of baths? Why or why not?

Submission

Upload three files to Canvas before the deadline:

- an R **script** (.R),
- your `week_08.Rmd`, and
- the **knitted PDF**.

As always: comment your code, and if you get stuck, show your work. If your .Rmd won't knit, get each piece working before you move on to the next, and restart R and knit clean if results look stale.